Exploratory Experiments on CNNs, ViT, CLIP, and VAEs

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Abstract

1. CNNs

1.1. Introduction

1.2. Methodology

1.3. Results

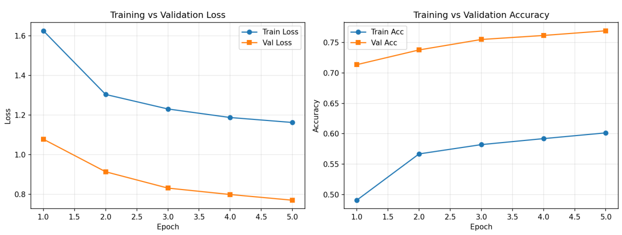


Figure 1. Training and validation loss of the fine-tuned ResNet-152 model over the training epochs.

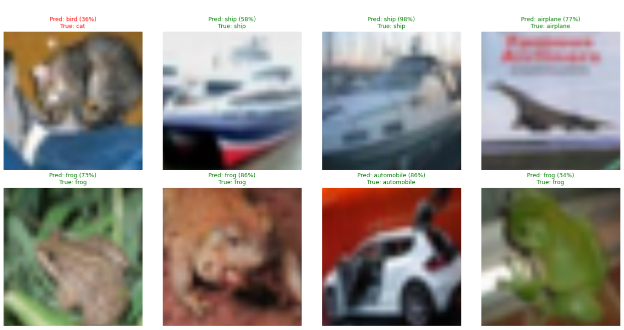


Figure 2. Example predictions from the fine-tuned ResNet-152 model on images from the evaluation set.

Table 1. Effect of patch masking on ViT predictions. Values show the confidence in the original prediction class under random and center masking. Bold values indicate higher confidence between the two masking strategies.

	Golden Retriever		**Sports Car**		**Egyptian Cat**	
Mask	Random	Center	Random	Center	Random	Center
0%	98.34	98.34	75.84	75.84	58.10	58.10
10%	98.57	97.55	85.34	75.96	59.40	70.74
20%	97.54	87.18	62.61	23.36	52.80	59.68
30%	97.98	61.63	78.69	23.92	55.35	58.68
40%	94.43	42.82	83.14	14.95	63.66	51.75
50%	97.29	47.89	77.39	20.19	39.95	60.85
60%	58.36	11.16	52.79	17.56	52.43	59.80
70%	94.77	3.53	61.00	17.08	48.81	48.13
80%	7.02	3.71	35.92	9.30	39.37	44.99

1.4. UMAP and t-SNE

1.5. Discussion

2. Vision Transformers (ViT)

2.1. Introduction

2.2. Methodology

2.3. Results

Table 2. Selected prediction changes under patch masking.

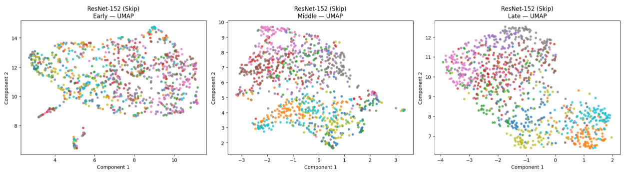
Image	**Mask**	**Random**	**Center**
Golden Retriever	50%	Golden Retriever	Matchstick
Golden Retriever	60%	Golden Retriever	Nematode
Golden Retriever	80%	Panpipe	Panpipe
Sports Car	60%	Convertible	Car mirror
Sports Car	70%	Car mirror	Car mirror
Egyptian Cat	50%	Tabby cat	Egyptian cat

2.4. Discussion

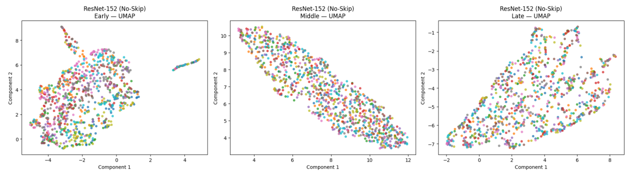
3. Contrastive Language-Image Pre-training (CLIP)

3.1. Introduction

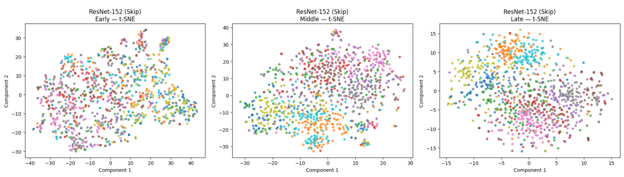
Contrastive Language–Image Pre-training (CLIP) is a multimodal representation learning framework introduced by OpenAI that learns a shared embedding space for images



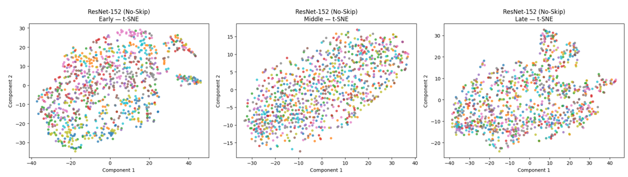
(a) UMAP with skip connections



(b) UMAP without skip connections



(c) t-SNE with skip connections



(d) t-SNE without skip connections

Figure 3. Comparison of learned feature representations using UMAP and t-SNE for ResNet-152 with and without skip connections. Each visualization shows features extracted from early, middle, and late layers of the network.

Figure 4. Sample images used in the ViT experiments.

Figure 5. Attention overlays from the final ViT layer, visualizing the image regions receiving the highest attention.

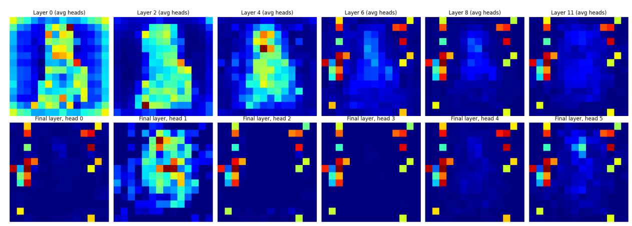


Figure 6. Attention patterns across different ViT layers for a dog image. The top row shows CLS-to-patch attention averaged across all heads for layers 0, 2, 4, 6, 8, and 11, illustrating the evolution of spatial attention throughout the network. The bottom row shows attention from six individual heads in the final layer, highlighting the diversity of attention patterns across heads.

and natural language. Instead of training directly on a fixed set of class labels, CLIP is pretrained on large-scale image–text pairs using a contrastive objective. Given an image and a collection of text descriptions, the model learns to increase the similarity between the corresponding image–text pair while decreasing the similarity to mismatched pairs.

This shared representation enables CLIP to perform zero-shot classification without requiring task-specific classifier heads. For example, an image can be compared against text prompts such as "a photo of a dog" or "a photo of a cat", with the class corresponding to the most similar text embedding selected as the prediction. This provides a flexible alternative to conventional supervised image classification, where a model is trained explicitly for a fixed set of classes.

The original CLIP study demonstrated that large-scale vision-language pretraining can produce strong zero-shot transfer performance across a wide range of datasets and tasks. Its ability to associate visual concepts with natural language also makes CLIP useful for applications where multiple modalities need to be represented within a common semantic space. The model consists of separate image and text encoders whose outputs are projected into the same embedding space, allowing their representations to be directly compared.

In this experiment, we investigate the behavior of CLIP under different prompting strategies for zero-shot classification. In particular, we compare plain class labels, simple natural-language templates, and more descriptive prompts. We further examine the structure of the resulting image and text embedding spaces and investigate whether Procrustes alignment can improve their correspondence.

Figure 7. Attention overlays from Layer 4 of the ViT. The visualization shows the spatial regions receiving attention from the CLS token after averaging attention across all heads.

Figure 8. Effect of patch masking on a ViT input image. The original image is compared with inputs where 30% of the image patches are masked either randomly or according to their proximity to the image center. Each masked patch is replaced with a zero-valued (black) region.

3.2. Methodology

Table 3. Prompt strategies with representative examples.

Strategy	**Example Prompt**
Plain labels	`airplane`
A photo of	`a photo of an airplane`
Descriptive	`a photo of an airplane in the sky`

3.3. Results

3.4. Discussion

4. Variational Autoencoders (VAE)

4.1. Introduction

4.2. Methodology

4.3. Results

4.4. Discussion

5. Conclusion

References

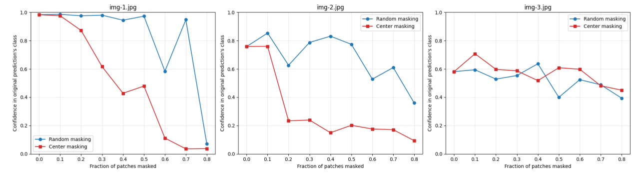


Figure 9. ViT confidence under increasing levels of patch masking. Each subplot corresponds to a different input image. Confidence is measured for the class predicted from the original unmasked image, while patches are progressively masked using either random or center-based masking.

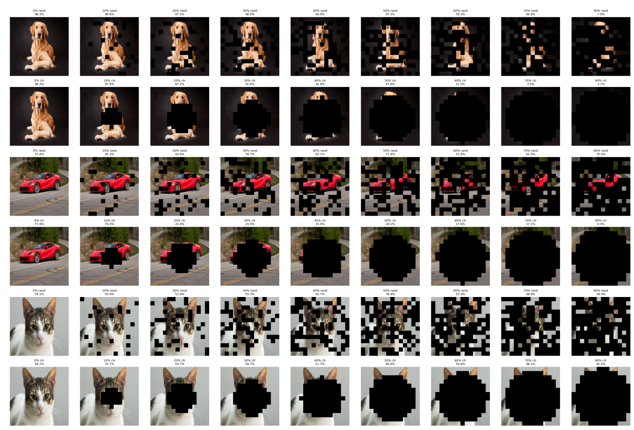


Figure 10. Qualitative effect of progressively masking ViT input patches. For each input image, the top row shows random masking and the bottom row shows center masking at increasing masking fractions. The confidence values indicate the model's confidence in the class predicted for the original unmasked image.

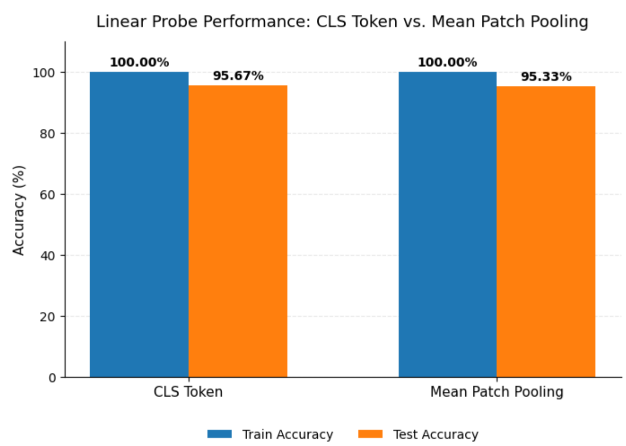


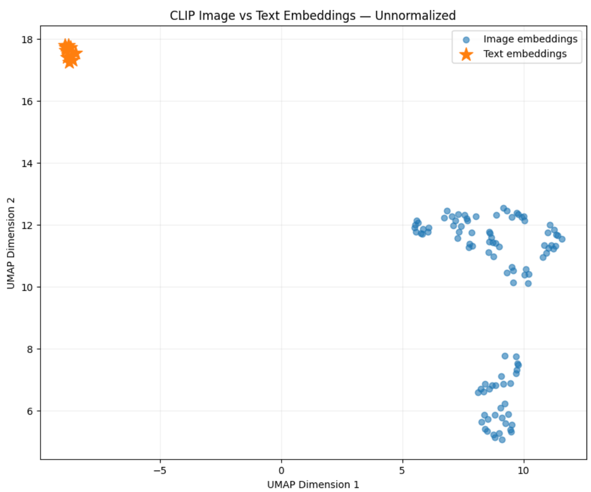
Figure 11. Comparison of linear probe performance using the final [CLS] token and mean-pooled patch token representations. Both representations achieve 100% training accuracy, while the [CLS] representation achieves slightly higher test accuracy (95.67%) than mean patch pooling (95.33%).

Table 4. Zero-shot classification accuracy using different text prompt formulations.

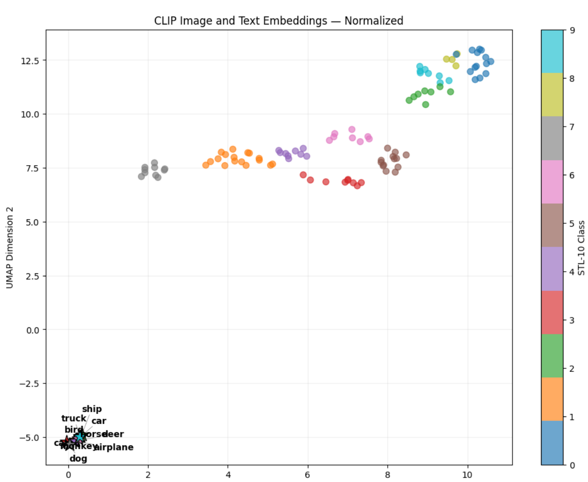
Prompt Type	**Accuracy (%)**
Plain labels	96.26
"A photo of"	**97.36**
Descriptive	90.80

Table 5. In-sample zero-shot classification accuracy before and after Procrustes alignment.

Method	**Accuracy (%)**
Original CLIP	97.36
After Procrustes	**100.00**
Change	**+2.64 pp**



(a) Before normalization



(b) After normalization

Figure 12. Visualization of the embedding space before and after normalization.

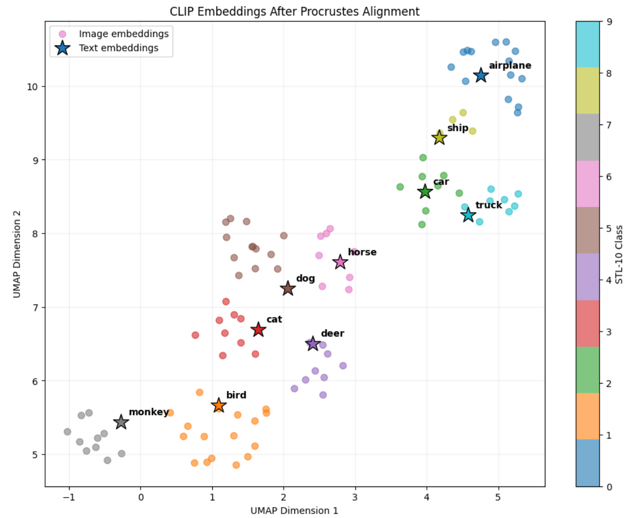


Figure 13. Visualization of CLIP embeddings after Procrustes alignment.

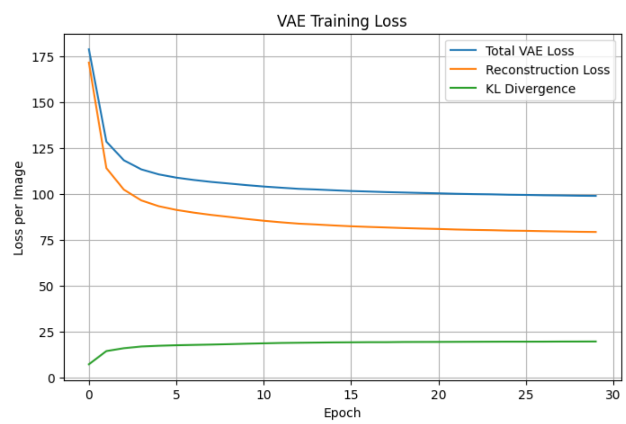


Figure 14. VAE training loss over training epochs. The loss captures the combined reconstruction and KL-divergence objectives used to optimize the variational autoencoder.

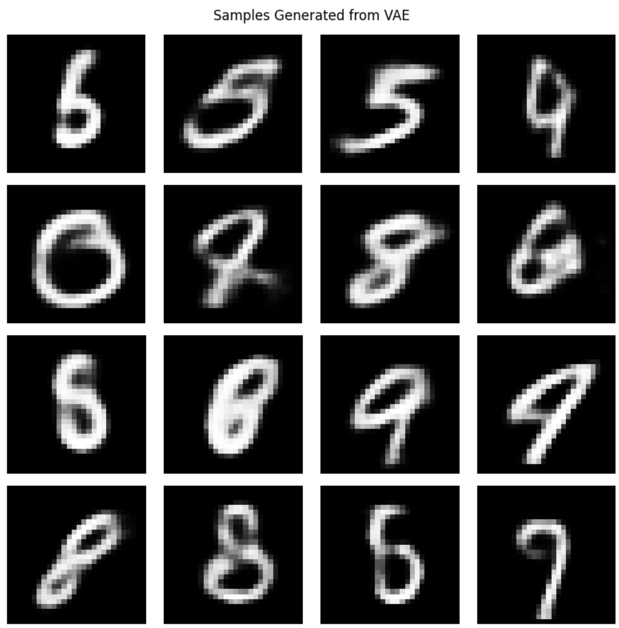


Figure 15. Samples generated by the trained VAE by decoding latent representations sampled from the learned latent distribution.

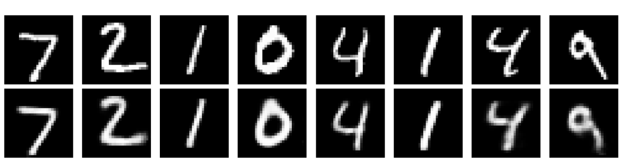


Figure 16. Original and reconstructed images produced by the trained VAE. The reconstructions illustrate how well the decoder preserves the salient visual features of the input images.

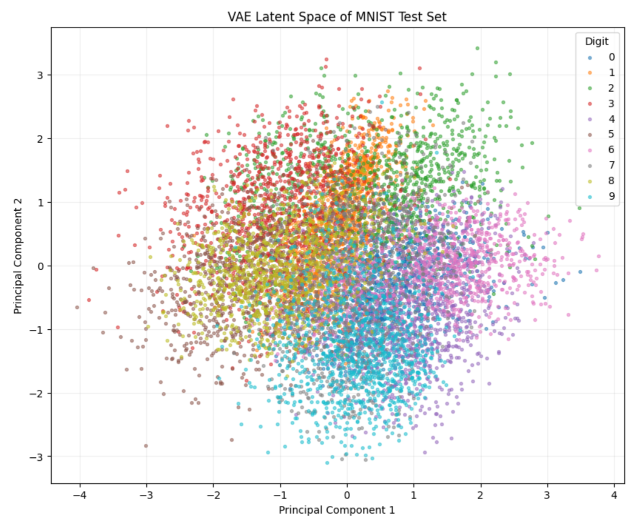


Figure 17. Two-dimensional latent space learned by the VAE on the MNIST test set. Each point represents the latent mean of a test image, with points colored according to their digit class.